

SQL Queries for the Chinook Database

Project Description

Chinook is a sample relational database that simulates the operation of a digital music store. It contains information about artists, albums, tracks, customers, employees, invoices, and sales.

As part of this project, a collection of SQL queries was developed to analyze data and solve various business scenarios. The project includes queries for data retrieval and filtering, table joins (JOIN), aggregation, ranking, as well as the calculation of key business metrics related to sales, revenue, and customer activity.

<https://github.com/lerocha/chinook-database>

SQL Queries

Customers

1. Get full user information by first name and last name

```
SELECT *
FROM Customer c
WHERE c.FirstName = 'Martha'
AND c.LastName = 'Silk';
```

123 Cust	AZ FirstN	AZ LastN	AZ Comp	AZ Addn	AZ City	AZ State	AZ Count	AZ Post	AZ Phon	AZ Fax	AZ Email	123 Supp
31	Martha	Silk	[NULL]	194A Chain L	Halifax	NS	Canada	B3S 1C5	+1 (902) 450-1	[NULL]	marthasilk@gr	5

2. Get full user information by email address

```
SELECT *
FROM Customer c
WHERE c.Email = 'puja_srivastava@yahoo.in'
```

123 Cust	AZ FirstN	AZ LastN	AZ Comp	AZ Addn	AZ City	AZ State	AZ Count	AZ Post	AZ Phon	AZ Fax	AZ Email	123 Supp
59	Puja	Srivastava	[NULL]	3,Raj Bhavan I	Bangalore	[NULL]	India	560001	+91 080 22285	[NULL]	puja_srivastav	3

3. Calculate the number of users by country

```

SELECT
Country,
Customers,
Percent
FROM
(SELECT c.Country,
COUNT (*) AS Customers,
ROUND(COUNT(*)*100.0/(SELECT COUNT(*) FROM Customer), 1) AS Percent,
0 AS SortOrder
FROM Customer c
GROUP BY c.Country
UNION ALL
SELECT 'TOTAL',
COUNT(*),
100.0,
1
FROM Customer)
ORDER BY SortOrder,
Customers DESC;

```

	 AZ Country ▼	123 Customers ▼	123 Percent ▼
1	USA	13	22
2	Canada	8	13.6
3	Brazil	5	8.5
4	France	5	8.5
5	Germany	4	6.8
6	United Kingdom	3	5.1
7	Czech Republic	2	3.4
8	India	2	3.4
9	Portugal	2	3.4
10	Argentina	1	1.7
11	Australia	1	1.7
12	Austria	1	1.7
13	Belgium	1	1.7
14	Chile	1	1.7
15	Denmark	1	1.7
16	Finland	1	1.7
17	Hungary	1	1.7
18	Ireland	1	1.7
19	Italy	1	1.7
20	Netherlands	1	1.7
21	Norway	1	1.7
22	Poland	1	1.7
23	Spain	1	1.7
24	Sweden	1	1.7
25	TOTAL	59	100

4. Calculate the number of users from company

SELECT COUNT (*) AS <i>fromCompany</i> , ROUND (COUNT (*)* 100.0 /(SELECT COUNT (*) FROM <i>Customer</i>), 1) AS <i>Percent</i> FROM <i>Customer</i> <i>c</i> WHERE <i>c.Company</i> IS NOT NULL ; 				
	123 <i>fromCompany</i> ▼	123 <i>Percent</i> ▼		
1	10	16.9		

5. Calculate the distribution of users among support staff

SELECT <i>SupportAgentId</i> , <i>AgentLastName</i> , <i>AgentFirstName</i> , <i>Customers</i> , <i>Percent</i> FROM (SELECT <i>c.SupportRepId</i> AS <i>SupportAgentId</i> , <i>e.LastName</i> AS <i>AgentLastName</i> , <i>e.FirstName</i> AS <i>AgentFirstName</i> , COUNT (*) AS <i>Customers</i> , ROUND (COUNT (*)* 100.0 /(SELECT COUNT (*) FROM <i>Customer</i>), 1) AS <i>Percent</i> , 0 AS <i>SortOrder</i> FROM <i>Customer</i> <i>c</i> INNER JOIN <i>Employee</i> <i>e</i> ON <i>e.EmployeeId</i> = <i>c.SupportRepId</i> GROUP BY <i>c.SupportRepId</i> , <i>e.LastName</i> , <i>e.FirstName</i> UNION ALL SELECT NULL , NULL , 'TOTAL', COUNT (*), 100.0 , 1 FROM <i>Customer</i>) ORDER BY <i>SortOrder</i> , <i>Customers</i> DESC ; 					
123 <i>SupportAgentId</i> ▼	AZ <i>AgentLastName</i> ▼	AZ <i>AgentFirstName</i> ▼	123 <i>Customers</i> ▼	123 <i>Percent</i> ▼	
3	Peacock	Jane	21	35.6	
4	Park	Margaret	20	33.9	
5	Johnson	Steve	18	30.5	
[NULL]	[NULL]	TOTAL	59	100	

Invoices

6. Get the selected user's purchase history

```
SELECT i.InvoiceId AS Id,
i.InvoiceDate AS DateOfInvoice,
i.Total AS InvoiceSum
FROM Invoice i
WHERE i.CustomerId = 30
ORDER BY Id;
```

	123 Id	A-Z DateOfInvoice	123 InvoiceSum
1	49	2021-08-06 00:00:00	1.98
2	72	2021-11-08 00:00:00	3.96
3	94	2022-02-10 00:00:00	5.94
4	146	2022-10-01 00:00:00	0.99
5	267	2024-03-24 00:00:00	1.98
6	278	2024-05-04 00:00:00	13.86
7	333	2025-01-02 00:00:00	8.91

7. Check if the user has purchased the same track more than once

```
SELECT il.TrackId,
SUM(il.Quantity) AS PurchaseCount
FROM Invoice i
INNER JOIN InvoiceLine il
ON il.InvoiceId = i.InvoiceId
WHERE i.CustomerId = 90
GROUP BY il.TrackId
HAVING SUM(il.Quantity) > 1;
```

	123 TrackId	123 PurchaseCount

8. Determine the top 20 users by total purchase amount

```
SELECT c.CustomerId,
c.FirstName,
c.LastName,
SUM(i.Total) AS TotalSpent
FROM Customer c
INNER JOIN Invoice i
ON c.CustomerId = i.CustomerId
GROUP BY c.CustomerId,
c.FirstName,
c.LastName
ORDER BY TotalSpent DESC
LIMIT 20;
```

	123 CustomerId	A-Z FirstName	A-Z LastName	123 TotalSpent
1	6	Helena	Holý	49.62
2	26	Richard	Cunningham	47.62
3	57	Luis	Rojas	46.62
4	45	Ladislav	Kovács	45.62
5	46	Hugh	O'Reilly	45.62
6	24	Frank	Ralston	43.62
7	28	Julia	Barnett	43.62
8	37	Fynn	Zimmermann	43.62
9	7	Astrid	Gruber	42.62
10	25	Victor	Stevens	42.62
11	44	Terhi	Hämäläinen	41.62
12	5	František	Wichterlová	40.62
13	43	Isabelle	Mercier	40.62
14	48	Johannes	Van der Berg	40.62
15	1	Luis	Gonçalves	39.62
16	3	François	Tremblay	39.62
17	4	Bjørn	Hansen	39.62
18	17	Jack	Smith	39.62
19	20	Dan	Miller	39.62
20	22	Heather	Leacock	39.62

9. Determine the company's average monthly revenue

```
SELECT ROUND(AVG(MonthRevenue), 2) AS AverageMonthlyRevenue
FROM
(SELECT strftime('%Y-%m', InvoiceDate) AS Month,
SUM(Total) AS MonthRevenue
FROM Invoice
GROUP BY strftime('%Y-%m', InvoiceDate));
```

123 AverageMonthlyRevenue
38.81

10. Determine the company's average revenue for each month across all years

```
SELECT m.Month,
ROUND(AVG(Revenue), 2) AS AverageRevenue
FROM
(SELECT strftime('%Y', i.InvoiceDate) AS Year,
strftime('%m', i.InvoiceDate) AS Month,
SUM(i.Total) AS Revenue
FROM Invoice i
GROUP BY Year,
Month) m
GROUP BY m.Month
ORDER BY m.Month;
```

AZ Month	123 AverageRevenue
01	40.22
02	37.44
03	39.02
04	39.63
05	38.62
06	40.22
07	38.02
08	39.62
09	39.24
10	38.62
11	37.25
12	37.82

11. Determine revenue by year

```
SELECT strftime('%Y', i.InvoiceDate) AS Year,  
ROUND(SUM(i.Total), 2) AS Revenue  
FROM Invoice i  
GROUP BY Year  
ORDER BY Year;
```

A-Z Year ▼	123 Revenue ▼
2021	449.46
2022	481.45
2023	469.58
2024	477.53
2025	450.58

12. Determine the revenue by month for the selected year

```
SELECT strftime('%m', i.InvoiceDate) AS Month,  
SUM(i.Total) AS Revenue  
FROM Invoice i  
WHERE strftime('%Y', i.InvoiceDate) = '2023'  
GROUP BY Month  
ORDER BY Month;
```

A-Z Month ▼	123 Revenue ▼
01	37.62
02	37.62
03	37.62
04	51.62
05	42.62
06	50.62
07	37.62
08	37.62
09	37.62
10	37.62
11	23.76
12	37.62

13. Determine the TOP 5 countries by total sales amount

```
SELECT i.BillingCountry AS Country,  
ROUND(SUM(i.Total), 2) AS Revenue  
FROM Invoice i  
GROUP BY Country  
ORDER BY Revenue DESC  
LIMIT 5;
```

	A-Z Country ▼	123 Revenue ▼
1	USA	523.06
2	Canada	303.96
3	France	195.1
4	Brazil	190.1
5	Germany	156.48

14. Determine the top 5 cities by total revenue

```
SELECT *  
FROM  
(SELECT i.BillingCity AS City,  
SUM(i.Total) AS Revenue,  
RANK() OVER (ORDER BY SUM(i.Total) DESC) AS Rank  
FROM Invoice i  
GROUP BY City) r  
WHERE r.Rank <= 5;
```

	A-Z City ▼	123 Revenue ▼	123 Rank ▼
1	Prague	90.24	1
2	Paris	77.24	2
3	Mountain View	77.24	2
4	São Paulo	75.24	4
5	London	75.24	4
6	Berlin	75.24	4

15. Determine the total revenue by genre

```
SELECT g.Name AS Genre,  
SUM(il.UnitPrice * il.Quantity) AS Revenue  
FROM Genre g  
INNER JOIN Track t  
ON t.GenreId = g.GenreId  
LEFT JOIN InvoiceLine il  
ON il.TrackId = t.TrackId  
GROUP BY g.Name  
ORDER BY Revenue DESC;
```

	A-Z Genre	123 Revenue
1	Rock	826.65
2	Latin	382.14
3	Metal	261.36
4	Alternative & Punk	241.56
5	TV Shows	93.53
6	Jazz	79.2
7	Blues	60.39
8	Drama	57.71
9	R&B/Soul	40.59
10	Classical	40.59
11	Sci Fi & Fantasy	39.8
12	Reggae	29.7
13	Pop	27.72
14	Soundtrack	19.8
15	Comedy	17.91
16	Hip Hop/Rap	16.83
17	Bossa Nova	14.85
18	Alternative	13.86
19	World	12.87
20	Science Fiction	11.94
21	Heavy Metal	11.88
22	Electronica/Dance	11.88
23	Easy Listening	9.9
24	Rock And Roll	5.94
25	Opera	[NULL]

Треки (Tracks)

16. Get full track's information by name

```
SELECT t.TrackId,
t.Name AS TrackName,
ar.Name AS Artist,
al.Title AS Album,
g.Name AS Genre,
t.Composer AS Composer,
mt.Name AS MediaType,
printf('%02d:%02d', t.Milliseconds / 60000, (t.Milliseconds % 60000) / 1000) AS
Duration,
ROUND(t.Bytes / 1024.0 / 1024.0, 2) AS SizeMB,
t.UnitPrice AS Price
FROM Track t
INNER JOIN Album al
ON al.AlbumId = t.AlbumId
INNER JOIN Artist ar
ON ar.ArtistId = al.ArtistId
INNER JOIN Genre g ON
g.GenreId = t.GenreId
INNER JOIN MediaType mt
ON mt.MediaTypeId = t.MediaTypeId
WHERE TrackName = 'Televisão';
```

123 TrackId	AZ TrackName	AZ Artist	AZ Album	AZ Genre	AZ Composer	AZ MediaType	AZ Duration	123 SizeMB	123 Price
2,802	Televisão	Titãs	Acústico	Alternative & Punk	Titãs	MPEG audio file	04:53	9.32	0.99

17. Determine the TOP 100 most popular tracks by sales

```
SELECT
t.TrackId,
t.Name AS TrackName,
SUM(il.Quantity) AS Sales
FROM InvoiceLine il
INNER JOIN Track t
ON il.TrackId = t.TrackId
GROUP BY t.TrackId
ORDER BY Sales DESC
LIMIT 100;
```

	123 TrackId	AZ TrackName	123 Sales
1	2	Balls to the Wall	2
2	8	Inject The Venom	2
3	9	Snowballed	2
4	20	Overdose	2
5	32	Deuces Are Wild	2
6	48	Not The Doctor	2
7	66	Por Causa De Você	2
8	84	Welcome Home (Sanita	2
9	161	Snowblind	2
10	162	Cornucopia	2
11	171	Bowels Of The Devil	2
12	175	Voodoo	2
13	184	Chemical Wedding	2
14	198	When My Left Eye Jump	2
15	207	Meditação	2
16	211	Bem Devagar	2

18. Determine the top 100 tracks by total revenue

```
SELECT
t.TrackId,
t.Name AS TrackName,
SUM(il.UnitPrice * il.Quantity) AS Revenue
FROM InvoiceLine il
INNER JOIN Track t
ON il.TrackId = t.TrackId
GROUP BY t.TrackId,
t.Name
ORDER BY Revenue DESC
LIMIT 100;
```

	123 TrackId	A-Z TrackName	123 Revenue
1	2,832	The Woman King	3.98
2	2,850	The Fix	3.98
3	2,868	Walkabout	3.98
4	3,177	Hot Girl	3.98
5	3,200	Gay Witch Hunt	3.98
6	3,214	Phyllis's Wedding	3.98
7	3,223	How to Stop an Explodi	3.98
8	3,250	Pilot	3.98
9	2,820	Occupation / Precipice	1.99
10	2,821	Exodus, Pt. 1	1.99
11	2,822	Exodus, Pt. 2	1.99
12	2,823	Collaborators	1.99
13	2,826	Hero	1.99
14	2,827	Unfinished Business	1.99
15	2,828	The Passage	1.99
16	2,833	A Day In the Life	1.99

19. Determine the number of unsold tracks

```
SELECT
COUNT(*) AS UnsoldTracks
FROM
(SELECT t.TrackId
FROM Track t
LEFT JOIN InvoiceLine il
ON t.TrackId = il.TrackId
GROUP BY t.TrackId
HAVING COUNT(il.InvoiceLineId) = 0);
```

123 UnsoldTracks
1,519

20. Determine the top 10 albums by number of tracks sold

```
SELECT
a.Title AS AlbumTitle,
SUM(il.Quantity) AS TotalSales
FROM InvoiceLine il
INNER JOIN Track t
ON il.TrackId = t.TrackId
INNER JOIN Album a
ON a.AlbumId = t.AlbumId
GROUP BY a.AlbumId,
a.Title
ORDER BY TotalSales DESC
LIMIT 10;
```

	  AZ AlbumTitle ▼	123 TotalSales ▼
1	Minha Historia	27
2	Greatest Hits	26
3	Unplugged	25
4	Acústico	22
5	Greatest Kiss	20
6	Prenda Minha	19
7	Chronicle, Vol. 2	19
8	My Generation - The Ve	19
9	International Superhits	18
10	Chronicle, Vol. 1	18

21. Determine the number of tracks by genre

```
SELECT g.Name AS Genre,  
COUNT(t.TrackId) AS TrackCount  
FROM Genre g  
INNER JOIN Track t  
ON t.GenreId = g.GenreId  
GROUP BY g.GenreId,  
g.Name  
ORDER BY TrackCount DESC;
```

	AZ Genre ▼	123 TrackCount ▼
1	Rock	1,297
2	Latin	579
3	Metal	374
4	Alternative & Punk	332
5	Jazz	130
6	TV Shows	93
7	Blues	81
8	Classical	74
9	Drama	64
10	R&B/Soul	61
11	Reggae	58
12	Pop	48
13	Soundtrack	43
14	Alternative	40
15	Hip Hop/Rap	35
16	Electronica/Dance	30
17	Heavy Metal	28
18	World	28
19	Sci Fi & Fantasy	26
20	Easy Listening	24
21	Comedy	17
22	Bossa Nova	15
23	Science Fiction	13
24	Rock And Roll	12
25	Opera	1

22. Determine the number of tracks sold by genre

```
SELECT g.Name AS Genre,  
SUM(il.Quantity) AS Sales  
FROM Genre g  
INNER JOIN Track t  
ON t.GenreId = g.GenreId  
LEFT JOIN InvoiceLine il  
ON il.TrackId = t.TrackId  
GROUP BY g.Name  
ORDER BY Sales DESC;
```

 	A-Z Genre ▼	123 Sales ▼
1	Rock	835
2	Latin	386
3	Metal	264
4	Alternative & Punk	244
5	Jazz	80
6	Blues	61
7	TV Shows	47
8	R&B/Soul	41
9	Classical	41
10	Reggae	30
11	Drama	29
12	Pop	28
13	Soundtrack	20
14	Sci Fi & Fantasy	20
15	Hip Hop/Rap	17
16	Bossa Nova	15
17	Alternative	14
18	World	13
19	Heavy Metal	12
20	Electronica/Dance	12
21	Easy Listening	10
22	Comedy	9
23	Science Fiction	6
24	Rock And Roll	6
25	Opera	[NULL]

Исполнители (Artists)

23. Get information about all tracks of an artist

```
SELECT t.Name AS Track,
t.TrackId AS ID,
g.Name AS Genre,
al.Title AS Album,
SUM(il.Quantity) AS Sales
FROM Artist ar
INNER JOIN Album al
ON al.ArtistId = ar.ArtistId
INNER JOIN Track t
ON al.AlbumId = t.AlbumId
LEFT JOIN InvoiceLine il
ON il.TrackId = t.TrackId
INNER JOIN Genre g
ON g.GenreId = t.GenreId
WHERE ar.ArtistId = 90
GROUP BY t.Name,
t.TrackId,
g.Name,
al.Title
ORDER BY al.Title,
t.TrackId;
```

 	A-Z Track ▾	123 ID ▾	A-Z Genre ▾	A-Z Album ▾	123 Sales ▾
1	Different World	1,201	Rock	A Matter of Life and Death	[NULL]
2	These Colours Do	1,202	Rock	A Matter of Life and Death	1
3	Brighter Than a Th	1,203	Rock	A Matter of Life and Death	1
4	The Pilgrim	1,204	Rock	A Matter of Life and Death	1
5	The Longest Day	1,205	Rock	A Matter of Life and Death	[NULL]
6	Out of the Shadow	1,206	Rock	A Matter of Life and Death	[NULL]
7	The Reincarnation	1,207	Rock	A Matter of Life and Death	[NULL]
8	For the Greater Go	1,208	Rock	A Matter of Life and Death	2
9	Lord of Light	1,209	Rock	A Matter of Life and Death	1
10	The Legacy	1,210	Rock	A Matter of Life and Death	[NULL]
11	Hallowed Be Thy I	1,211	Rock	A Matter of Life and Death	[NULL]
12	The Number Of Th	1,212	Metal	A Real Dead One	[NULL]
13	The Trooper	1,213	Metal	A Real Dead One	1
14	Prowler	1,214	Metal	A Real Dead One	1
15	Transylvania	1,215	Metal	A Real Dead One	1
16	Remember Tomor	1,216	Metal	A Real Dead One	[NULL]

24. Determine the top 10 artists by number of tracks released

```
SELECT ar.Name AS Artist,  
ar.ArtistId AS ID,  
COUNT(t.TrackId) AS Tracks  
FROM Artist ar  
INNER JOIN Album al  
ON al.ArtistId = ar.ArtistId  
INNER JOIN Track t  
ON al.AlbumId = t.AlbumId  
GROUP BY ar.Name,  
ar.ArtistId  
ORDER BY Tracks DESC  
LIMIT 10;
```

	  AZ Artist ▼	123  ID ▼	123 Tracks ▼
1	Iron Maiden	90	213
2	U2	150	135
3	Led Zeppelin	22	114
4	Metallica	50	112
5	Deep Purple	58	92
6	Lost	149	92
7	Pearl Jam	118	67
8	Lenny Kravitz	100	57
9	Various Artists	21	56
10	The Office	156	53

25. Determine the top 10 artists by number of tracks sales

```
SELECT ar.Name AS Artist,  
ar.ArtistId AS ID,  
SUM(il.Quantity) AS Sales,  
COUNT(DISTINCT t.TrackId) AS Tracks  
FROM Artist ar  
INNER JOIN Album al  
ON al.ArtistId = ar.ArtistId  
INNER JOIN Track t  
ON al.AlbumId = t.AlbumId  
LEFT JOIN InvoiceLine il  
ON il.TrackId = t.TrackId  
GROUP BY ar.Name,  
ar.ArtistId  
ORDER BY Sales DESC  
LIMIT 10;
```

	A-Z Artist ▼	123 ID ▼	123 Sales ▼	123 Tracks ▼
1	Iron Maiden	90	140	213
2	U2	150	107	135
3	Metallica	50	91	112
4	Led Zeppelin	22	87	114
5	Os Paralamas Do	113	45	49
6	Deep Purple	58	44	92
7	Faith No More	82	42	52
8	Lost	149	41	92
9	Eric Clapton	81	40	48
10	R.E.M.	124	39	41

26. Determine the top 10 artists by number of unique customers

```

SELECT *
FROM
(SELECT ar.Name AS Artist,
ar.ArtistId AS ID,
RANK() OVER(ORDER BY COUNT(DISTINCT i.CustomerId) DESC) AS ArtistRank,
COUNT(DISTINCT i.CustomerId) AS Customers
FROM Artist ar
INNER JOIN Album al
ON al.ArtistId = ar.ArtistId
INNER JOIN Track t
ON al.AlbumId = t.AlbumId
INNER JOIN InvoiceLine il
ON il.TrackId = t.TrackId
INNER JOIN Invoice i
ON i.InvoiceId = il.InvoiceId
GROUP BY ar.Name,
ar.ArtistId) r
WHERE r.ArtistRank <= 10
ORDER BY Customers DESC;

```

	 	AZ Artist ▼	123 ID ▼	123 ArtistRank ▼	123 Customers ▼
1		U2	150	1	29
2		Led Zeppelin	22	2	28
3		Iron Maiden	90	3	27
4		Metallica	50	3	27
5		Eric Clapton	81	5	16
6		Os Paralamas Do	113	5	16
7		Lenny Kravitz	100	7	15
8		Queen	51	7	15
9		R.E.M.	124	7	15
10		Green Day	54	10	14
11		Various Artists	21	10	14

27. Determine the total revenue generated by each artist

```
SELECT ar.Name AS Artist,
ar.ArtistId AS ID,
SUM(il.UnitPrice) AS Revenue,
ROUND(SUM(il.UnitPrice)*100/(SELECT SUM(UnitPrice * Quantity) FROM InvoiceLine), 2)
AS RevenuePercentage,
COUNT(DISTINCT t.TrackId) AS Tracks
FROM Artist ar
LEFT JOIN Album al
ON al.ArtistId = ar.ArtistId
LEFT JOIN Track t
ON al.AlbumId = t.AlbumId
LEFT JOIN InvoiceLine il
ON il.TrackId = t.TrackId
GROUP BY ar.Name,
ar.ArtistId
ORDER BY Revenue DESC;
```

	A-Z Artist	123 ID	123 Revenue	123 RevenuePercentage	123 Tracks
1	Iron Maiden	90	138.6	5.95	213
2	U2	150	105.93	4.55	135
3	Metallica	50	90.09	3.87	112
4	Led Zeppelin	22	86.13	3.7	114
5	Lost	149	81.59	3.5	92
6	The Office	156	49.75	2.14	53
7	Os Paralamas Do	113	44.55	1.91	49
8	Deep Purple	58	43.56	1.87	92
9	Faith No More	82	41.58	1.79	52
10	Eric Clapton	81	39.6	1.7	48
11	R.E.M.	124	38.61	1.66	41
12	Queen	51	36.63	1.57	45
13	Creedence Clearw	76	36.63	1.57	40
14	Battlestar Galactic	158	35.82	1.54	24
15	Guns N' Roses	88	35.64	1.53	42
16	Titãs	146	33.66	1.45	38

28. Determine the most popular artist in each country

```

SELECT r.Country, r.Artist, r.ID, r.Sales, r.Tracks
FROM
(SELECT i.BillingCountry AS Country,
ar.Name AS Artist,
ar.ArtistId AS ID,
SUM(il.Quantity) AS Sales,
(SELECT COUNT(*)
FROM Album a2
INNER JOIN Track t2
ON t2.AlbumId = a2.AlbumId
WHERE a2.ArtistId = ar.ArtistId) AS Tracks,
RANK() OVER (PARTITION BY i.BillingCountry ORDER BY SUM(il.Quantity) DESC) AS
ArtistRank
FROM Artist ar
INNER JOIN Album al
ON al.ArtistId = ar.ArtistId
INNER JOIN Track t
ON al.AlbumId = t.AlbumId
INNER JOIN InvoiceLine il
ON il.TrackId = t.TrackId
INNER JOIN Invoice i
ON i.InvoiceId = il.InvoiceId
GROUP BY i.BillingCountry, ar.Name, ar.ArtistId) r
WHERE r.ArtistRank = 1
ORDER BY Sales DESC;

```

	AZ Country	AZ Artist	123 ID	123 Sales	123 Tracks
1	USA	Iron Maiden	90	34	213
2	Australia	Iron Maiden	90	18	213
3	Canada	Os Paralamas Do	113	16	49
4	Portugal	Iron Maiden	90	16	213
5	Germany	Iron Maiden	90	14	213
6	Brazil	Os Paralamas Do	113	11	49
7	Brazil	Pearl Jam	118	11	67
8	France	Creedence Clearw	76	11	40
9	Austria	U2	150	9	135
10	Belgium	Faith No More	82	9	52
11	Czech Republic	Iron Maiden	90	9	213
12	Czech Republic	U2	150	9	135
13	Norway	Led Zeppelin	22	9	114
14	United Kingdom	Metallica	50	9	112
15	India	Iron Maiden	90	8	213
16	Ireland	Lost	149	8	92
17	Denmark	Creedence Clearw	76	7	40
18	Argentina	Metallica	50	6	112
19	Chile	Led Zeppelin	22	6	114
20	Finland	Van Halen	152	6	52
21	Netherlands	Red Hot Chili Pep	127	6	48
22	Poland	U2	150	6	135
23	Spain	Pearl Jam	118	6	67
24	Hungary	Jamiroquai	92	5	32
25	Hungary	The Office	156	5	53
26	Italy	Faith No More	82	5	52
27	Italy	The Rolling Stone	142	5	41
28	Sweden	Titäs	146	5	38

29. Determine the artists with the most sales for each year

```
SELECT r.Year,
r.Artist,
r.ID,
r.Sales,
r.Tracks
FROM
(SELECT strftime('%Y', i.InvoiceDate) AS Year,
ar.Name AS Artist,
ar.ArtistId AS ID,
SUM(il.Quantity) AS Sales,
COUNT(DISTINCT t.TrackId) AS Tracks,
RANK() OVER (PARTITION BY strftime('%Y', i.InvoiceDate) ORDER BY SUM(il.Quantity)
DESC) AS ArtistRank
FROM Artist ar
INNER JOIN Album al
ON al.ArtistId = ar.ArtistId
INNER JOIN Track t
ON al.AlbumId = t.AlbumId
INNER JOIN InvoiceLine il
ON il.TrackId = t.TrackId
INNER JOIN Invoice i
ON i.InvoiceId = il.InvoiceId
GROUP BY Year,
ar.Name,
ar.ArtistId) r
WHERE ArtistRank = 1
ORDER BY r.Year DESC;
```

AZ Year ▼	AZ Artist ▼	123 ID ▼	123 Sales ▼	123 Tracks ▼	
2025	Iron Maiden	90	36	36	
2024	Iron Maiden	90	34	34	
2023	U2	150	27	27	
2022	Iron Maiden	90	35	35	
2021	Iron Maiden	90	34	34	